

SUSHANTH TIRUVAIPATI

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EXPERIENCE

Software Engineer · Google

Sep 2015 – Present (10+ years)

Delivering high-impact projects across Ads, Cloud, Play, and YouTube, with experience leading teams and mentoring engineers.

Ads — Dynamic Display Ads & Conversion Tracking [C++ · Python · SQL · TensorFlow · Spanner · Borg]

- Lead conversion attribution systems that model how users interact with ads across Shopping, Search, Gmail, and YouTube to improve ad relevance and bidding accuracy.
- Drove unification of conversion tracking signals across Google's ad surfaces, enabling consistent measurement and attribution at scale.
- Led a team of 5 engineers researching proprietary and experimental user-attribution methodologies.
- Ensured ad systems remained compliant with evolving privacy regulations including the EU Digital Markets Act (DMA).
- 20% project: actor-safety enforcement system to detect and remove fraudulent advertisers.
- 20% project: latency improvements for Google Pay transaction flows.

Google Cloud — Contact Center AI (CCAI) [Java · Python · TensorFlow · NLP · gRPC]

- Part of the Cloud AI organization building AI-powered vertical solutions for enterprise customers.
- Contributed to Contact Center AI, a platform transforming the contact-center industry with conversational AI and NLU.
- Built topic-modeling solutions that automatically extract and summarize key topics from customer conversations at scale.

Google Play — Search Ranking [C++ · Python · TensorFlow · MapReduce · A/B experimentation]

- Worked on prefix search ranking for the Google Play Store, improving relevance and quality of real-time query results.
- Developed and iterated on ranking models balancing user-intent signals, app quality, and engagement metrics to surface the most relevant apps.
- Contributed to ranking-model evaluation pipelines and A/B experimentation frameworks measuring impact on engagement and install rates.

YouTube — Data & ML (Promo Performance) [Java · Python · SQL · Hadoop · MapReduce · Dataflow]

- Built and maintained large-scale data pipelines for analyzing promotional campaign performance across YouTube's advertising surfaces.
- Delivered reliable, low-latency datasets tracking promo reach, engagement, and conversion metrics for cross-functional teams.
- Collaborated with ML and product teams to ensure pipeline outputs fed downstream models and dashboards driving campaign optimization decisions.

PERSONAL PROJECTS & SIDE WORK

Founder & Sole Engineer · **KreativeKoala Solutions LLC**

2021 – Present

Solo-operator mobile portfolio of 70+ shipped apps with active users and revenue across App Store and Google Play.

Mobile portfolio & AI integration [Swift · Kotlin · TypeScript · Node.js · Python]

- Designed, built, and operate 70+ iOS / Android apps end-to-end — UI, backend, payments, ASO, App Store / Play Store compliance, growth analytics.
- Shipped deep LLM integration (Anthropic, OpenAI) into multiple consumer apps — Audexa (audio briefings), ScribeAI (study tool with 17K+ organic monthly installs), MeetingMind (meeting capture + summarization).
- Operate a unified PaywallKit + RatingKit cross-app SDK with server-driven A/B testing, used across the portfolio for monetization and engagement.

AI agent infrastructure (VibeBuild) [Node.js · TypeScript · Python · Kubernetes · Ray · Docker · Bash · Cloud Run]

- Built and operate a production AI app-generation pipeline: orchestrates Claude Code CLI as a build engine, spawning sandboxed per-project workers with OAuth credential rotation and quota management.
- Migrated the worker fleet from bare-metal PM2 to managed Kubernetes (GKE / Hetzner) with HPA based on queue depth — ~30% lower idle cost, zero-downtime deploys.
- Parallelized AI generation with Ray actors across cluster nodes, scaling concurrent Claude-CLI invocations from 4 to 32+ with shared scheduling and live dashboard.
- Hands-on experience running an AI coding agent as production infrastructure — sandboxing, distributed scheduling, observability — the same shape of problem agent-infra labs are working on.

EARLIER EXPERIENCE

Member of Technical Staff · **VMware**

Mar 2015 – Sep 2015

- vSphere product installer and cross-system compatibility. [C++ · Python]

Software Development Engineer · **Microsoft**

Nov 2012 – Feb 2015

- Office 365: scalable cloud-based infrastructure processing large volumes of customer data. Enabled enterprise adoption of modern productivity apps. [C# · C++ · Azure · PowerShell]

Software Development Engineer · **Amazon**

Oct 2011 – Oct 2012

- AWS: built robust billing and marketplace systems handling high transaction volumes; designed for future scale. [Java · AWS · DynamoDB · SQS]

EDUCATION

Carnegie Mellon University

M.S., Information Networking · 2009 – 2011

Indian Institute of Information Technology

B.Tech., Information Technology · 2004 – 2008

SKILLS

Languages: Python · Java · Swift · Kotlin · TypeScript / JavaScript · C / C++ · SQL · Bash · Go (working) · Rust (ramping)

AI / ML: LLM integration (Anthropic, OpenAI) · TensorFlow · NLP · Ranking models · Recommendation systems · Ray

Infrastructure: Kubernetes (GKE, Hetzner-managed) · Docker · Google Cloud (Cloud Run, Cloud Build, Spanner, Pub/Sub) · AWS · Hadoop · MapReduce · Dataflow · gRPC

Mobile / Product: iOS (Swift, SwiftUI, WidgetKit, ActivityKit) · Android (Kotlin, Jetpack Compose) · App Store / Play Store distribution · StoreKit 2 · Firebase

Other: Distributed systems · Data pipelines · A/B experimentation · Privacy & compliance (DMA, GDPR) · Team lead (5 engineers)